33.

Compound	Given no. of moles	No. of oxygen atoms
Ethanol - C_2H_5OH	1	$1 \times 6.022 \times 10^{23}$
Formic acid - HCOOH	1	$2 \times 6.022 \times 10^{23}$
Water - H_2O	1	$1 \times 6.022 \times 10^{23}$
Answer : Formic acid		

34. Average atomic mass $= \frac{(78.99 \times 23.99) + (10 \times 24.99) + (11.01 \times 25.98)}{100}$

$= \frac{2430.9}{100}$

$= 24.31u$

35. Reaction : $x + y + z_2 \rightarrow xyz_2$

Question	Number of moles of reactants allowed to react			Number of moles of reactants consumed during reaction			Limiting reagent
	x	y	z_2	x	y	z_2	
(a)	200 atoms	200 atoms	50 molecules	50 atoms	50 atoms	50 molecules	z_2
(b)	1 mol	1mol	3 mol	1 mol	1 mol	1 mol	x and y
(c)	50 atom	25 atom	50 molecules	25 atom	25 atom	25 molecules	y
(d)	2.5 mol	5 mol	5 mol	2.5 mol	2.5 mol	2.5 mol	x

36. Given: mass of one atom $= 6.645 \times 10^{-23}$ g

$\therefore$ mass of 1 mole of atom $= 6.645 \times 10^{-23}$ g $\times 6.022 \times 10^{23}$

$= 40$ g

$\therefore$ number of moles of element in 0.320 kg $= \frac{1 \text{ mole}}{40 \text{ g}} \times 0.320 \text{ kg}$

$= \frac{1 \text{ mol} \times 320 \text{ g}}{40 \text{ g}}$

$= 8$ mol.

38.

Compound	Molecular formula	Empirical formula
Fructose	$C_6H_{12}O_6$	CH_2O
Caffeine	$C_8H_{10}N_4O_2$	$C_4H_5N_2O$

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